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## TREND-BASED MOCK PAPER

### GATE Metallurgical Engineering

#### Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

#### Examination Details

|                   |                           |
|-------------------|---------------------------|
| Branch:           | Metallurgical Engineering |
| Paper:            | Mock 04                   |
| Total Questions:  | 65                        |
| Total Marks:      | 100                       |
| Duration:         | 3 Hours                   |
| Question Types:   | MCQ, MSQ, NAT             |
| Negative Marking: | Yes (MCQ only)            |
| Difficulty:       | Mixed                     |

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Historical patterns do not guarantee future examination questions.  
This is a trend-based practice simulation.

## GENERAL INSTRUCTIONS

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1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
  - MCQ (Single Correct Answer): 1 or 2 marks each
  - MSQ (Multiple Correct Answers): 2 marks each
  - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.  
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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## Section A: General Aptitude

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### Q1. [1-MCQ] [1 mark]

10% of 500:

- (A) 5
- (B) 50
- (C) 500
- (D) 0.5

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## Section B: Metallurgical Engineering

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### Q2. [1-MCQ] [1 mark]

Rank of 3x3 zero matrix:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

---

### Q3. [1-MCQ] [1 mark]

If  $f(x) = x^3 - 3x + 2$ , number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

---

### Q4. [1-MCQ] [1 mark]

For the ODE  $dy/dx = y/x + \sqrt{y^2/x^2}$ , with  $y(1)=0$ , the solution is:

- (A)  $y = 0$
- (B)  $y = x \ln(x)$
- (C)  $y = x \sin(\ln x)$

(D)  $y = x^2$

---

**Q5. [1-MCQ] [1 mark]**

Cauchy-Schwarz inequality states  $|\langle u, v \rangle| \leq$ :

- (A)  $\|u\| * \|v\|$
- (B)  $\|u\| + \|v\|$
- (C)  $\|u\| / \|v\|$
- (D)  $\sqrt{\|u\| * \|v\|}$

---

**Q6. [1-MCQ] [1 mark]**

Laplace transform of  $e^{-3t}u(t)$  is:

- (A)  $1/(s+3)$
- (B)  $1/(s-3)$
- (C)  $1/s$
- (D)  $e^{-3}/s$

---

**Q7. [1-MCQ] [1 mark]**

Rolle's theorem requires  $f(x)$  to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on  $[a, b]$  and differentiable on  $(a, b)$
- (D) Both continuous and differentiable everywhere

---

**Q8. [1-MCQ] [1 mark]**

$d/dx[\ln(\sec x) + \ln(\tan x)]$  evaluated at  $x = \pi/4$  equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

---

**Q9. [1-MCQ] [1 mark]**

Number of spanning trees in complete graph  $K_4$ :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

---

**Q10. [1-MCQ] [1 mark]**

$P(A \cup B) = 0.6$ ,  $P(A \cap B) = 0.2$ ,  $P(B) = 0.4$ . Then  $P(A|B)$  equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

---

**Q11. [1-MCQ] [1 mark]**

Fourier series of  $f(x)=x$  on  $(-\pi, \pi)$  has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

---

**Q12. [1-MCQ] [1 mark]**

The rank of the matrix  $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$  is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

---

**Q13. [1-NAT] [1 mark]**

If  $f(z) = z^2 + 3z + 2$  is analytic, then  $\operatorname{Re}(f(i)) =$

*(Numerical Answer Type — Enter the numerical value)*

---

**Q14. [2-MCQ] [2 marks]**

$P(|Z| < 1)$  for  $Z \sim N(0,1)$  is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

---

**Q15. [2-MCQ] [2 marks]**

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

---

**Q16. [2-MCQ] [2 marks]**

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

---

**Q17. [2-NAT] [2 marks]**

If a  $3 \times 3$  matrix has eigenvalues 1, 2, and 3, then  $\operatorname{trace}(A^2)$  equals:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q18. [2-MCQ] [2 marks]**

For matrix  $A = \begin{bmatrix} 2,1 \\ 1,2 \end{bmatrix}$ ,  $A^{50} \cdot \begin{bmatrix} 1,0 \end{bmatrix}^T$  equals:

- (A)  $\begin{bmatrix} 1,0 \end{bmatrix}^T$
- (B)  $\begin{bmatrix} 2,1 \end{bmatrix}^T$
- (C)  $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

---

**Q19. [2-NAT] [2 marks]**

Which of the following is a probability generating function for  $\operatorname{Poisson}(\lambda)$ ?

*(Numerical Answer Type — Enter the numerical value)*

---

**Q20. [2-MCQ] [2 marks]**

The differential equation  $y \, dx - x \, dy = 0$  has solution:

- (A)  $y = C \cdot x$

- (B)  $y = C/x$
- (C)  $y = C \cdot x^2$
- (D)  $x^2 + y^2 = C$

---

**Q21. [2-MCQ] [2 marks]**

The vector (1,2,3) projected onto (2,1,0) is:

- (A) (4/5,2/5,0)
- (B) (8/5,4/5,0)
- (C) (2,1,0)
- (D) (1,2,3)

---

**Q22. [2-NAT] [2 marks]**

The Laplace transform of  $t \cdot e^{-2t}$  is:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q23. [2-MCQ] [2 marks]**

The area of a circle of radius  $r$  is:

- (A)  $\pi \cdot r^2$
- (B)  $2 \cdot \pi \cdot r$
- (C)  $\pi \cdot r$
- (D)  $2 \cdot \pi \cdot r^2$

---

**Q24. [2-MCQ] [2 marks]**

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$  equals:

- (A)  $e$
- (B)  $e^2$
- (C) 1
- (D) infinity

---

**Q25. [2-MCQ] [2 marks]**

A random variable  $X \sim \text{Poisson}(\lambda=4)$ .  $P(X \geq 2)$  is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

---

**Q26. [2-NAT] [2 marks]**

Consider  $f(x) = |x|$  on  $[-1,1]$ . This function is:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q27. [2-MCQ] [2 marks]**

Solution of  $y'' + 4y = 0$  with  $y(0)=1$ ,  $y'(0)=0$ ,  $y(\pi/2)$  equals:

- (A) 1
- (B) 0
- (C) -1
- (D)  $\cos(\pi)$

---

**Q28. [2-NAT] [2 marks]**

The rank of the matrix  $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$  is:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q29. [2-NAT] [2 marks]**

The vector (1,2,3) projected onto (2,1,0) is:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q30. [2-MCQ] [2 marks]**

$d/dx[\ln(\sec x) + \ln(\tan x)]$  evaluated at  $x = \pi/4$  equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

---

**Q31. [1-MCQ] [1 mark]**

BCC coordination number:

- (A) 4
- (B) 6
- (C) 8
- (D) 12

---

**Q32. [1-MCQ] [1 mark]**

Phase diagram lever rule finds:

- (A) Temperature
- (B) Phase fraction
- (C) Composition
- (D) Density

---

**Q33. [1-NAT] [1 mark]**

Eutectic reaction:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q34. [1-NAT] [1 mark]**

BCC structure coordination number:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q35. [2-MCQ] [2 marks]**

Eutectic reaction:

- (A) Liquid  $\rightarrow$   $\alpha$  +  $\beta$
- (B)  $\alpha$   $\rightarrow$   $\beta$  +  $\gamma$
- (C) Liquid  $\rightarrow$   $\alpha$
- (D)  $\alpha$  +  $\beta$   $\rightarrow$   $\gamma$

---

**Q36. [2-MSQ] [2 marks]**

BCC structure coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

---

**Q37. [2-MCQ] [2 marks]**

Phase diagram lever rule finds:

- (A) Temperature

- (B) Phase fraction
- (C) Composition
- (D) Density

---

**Q38. [2-MSQ] [2 marks]**

BCC structure coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

---

**Q39. [2-MCQ] [2 marks]**

BCC structure coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

---

**Q40. [2-MSQ] [2 marks]**

Eutectic reaction:

- (A) Liquid  $\rightarrow$  alpha + beta
- (B) alpha  $\rightarrow$  beta + gamma
- (C) Liquid  $\rightarrow$  alpha
- (D) alpha + beta  $\rightarrow$  gamma

---

**Q41. [2-NAT] [2 marks]**

Eutectic reaction:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q42. [1-MCQ] [1 mark]**

BCC coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

---

**Q43. [1-MCQ] [1 mark]**

Brass is:

- (A) Cu-Zn
- (B) Cu-Sn
- (C) Cu-Ni
- (D) Cu-Al

---

**Q44. [1-NAT] [1 mark]**

FCC coordination number:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q45. [1-MCQ] [1 mark]**

Steel hardening process:

- (A) Annealing
- (B) Quenching

- (C) Normalizing
- (D) Tempering

---

**Q46. [2-NAT] [2 marks]**

FCC coordination number:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q47. [2-MCQ] [2 marks]**

BCC coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

---

**Q48. [2-NAT] [2 marks]**

Steel hardening process:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q49. [2-MCQ] [2 marks]**

FCC coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

---

**Q50. [2-NAT] [2 marks]**

BCC coordination number:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q51. [2-MCQ] [2 marks]**

Steel hardening process:

- (A) Annealing
- (B) Quenching
- (C) Normalizing
- (D) Tempering

---

**Q52. [1-MCQ] [1 mark]**

Reducing agent in blast furnace:

- (A) Coke
- (B) Limestone
- (C) Hot air
- (D) Slag

---

**Q53. [1-MCQ] [1 mark]**

Halls-Heroult process produces:

- (A) Iron
- (B) Aluminum
- (C) Copper
- (D) Zinc

---

**Q54. [1-NAT] [1 mark]**

Blast furnace main product:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q55. [2-NAT] [2 marks]**

Halls-Heroult process produces:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q56. [2-MSQ] [2 marks]**

Halls-Heroult process produces:

- (A) Iron
  - (B) Aluminum
  - (C) Copper
  - (D) Zinc
- 

**Q57. [2-MCQ] [2 marks]**

Halls-Heroult process produces:

- (A) Iron
  - (B) Aluminum
  - (C) Copper
  - (D) Zinc
- 

**Q58. [2-MCQ] [2 marks]**

Blast furnace main product:

- (A) Steel
  - (B) Pig iron
  - (C) Copper
  - (D) Aluminum
- 

**Q59. [2-NAT] [2 marks]**

Blast furnace main product:

*(Numerical Answer Type — Enter the numerical value)*

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**Q60. [1-NAT] [1 mark]**

Edge dislocation moves by:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q61. [1-MCQ] [1 mark]**

Eutectic point has:

- (A) One phase
  - (B) Two phases
  - (C) Three phases simultaneously
  - (D) Four phases
- 

**Q62. [1-NAT] [1 mark]**

Martensite is formed by:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q63. [1-MCQ] [1 mark]**

Cathodic reaction in rusting:

- (A)  $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$
- (B)  $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$
- (C) Both
- (D) None

---

**Q64. [1-NAT] [1 mark]**

Shielded metal arc uses:

*(Numerical Answer Type — Enter the numerical value)*

---

**Q65. [1-MCQ] [1 mark]**

Duralumin is Al +:

- (A) Cu
  - (B) Mg
  - (C) Si
  - (D) Mn
-

## ANSWER KEY

| Q. No. | Type | Answer | Marks |
|--------|------|--------|-------|
| 1      | 1MCQ | B      | 1     |
| 2      | 1MCQ | A      | 1     |
| 3      | 1MCQ | C      | 1     |
| 4      | 1NAT | B      | 1     |
| 5      | 1MCQ | A      | 1     |
| 6      | 1MCQ | C      | 1     |
| 7      | 1NAT | B      | 1     |
| 8      | 1MCQ | B      | 1     |
| 9      | 1NAT | B      | 1     |
| 10     | 1MCQ | A      | 1     |
| 11     | 1MCQ | B      | 1     |
| 12     | 1NAT | A      | 1     |
| 13     | 1NAT | B      | 1     |

|    |      |   |   |
|----|------|---|---|
| 14 | 1MCQ | B | 1 |
| 15 | 1MCQ | B | 1 |
| 16 | 1MCQ | A | 1 |
| 17 | 1MCQ | A | 1 |
| 18 | 1NAT | C | 1 |
| 19 | 1MCQ | A | 1 |
| 20 | 1MCQ | B | 1 |
| 21 | 1MCQ | B | 1 |
| 22 | 1NAT | B | 1 |
| 23 | 1MCQ | A | 1 |
| 24 | 1MCQ | C | 1 |
| 25 | 1MCQ | C | 1 |
| 26 | 1MCQ | C | 1 |
| 27 | 1MCQ | B | 1 |
| 28 | 1MCQ | A | 1 |

|    |      |   |   |
|----|------|---|---|
| 29 | 1MCQ | A | 1 |
| 30 | 1NAT | A | 1 |
| 31 | 2MCQ | A | 2 |
| 32 | 2MCQ | C | 2 |
| 33 | 2MCQ | A | 2 |
| 34 | 2NAT | A | 2 |
| 35 | 2MCQ | A | 2 |
| 36 | 2NAT | A | 2 |
| 37 | 2MCQ | B | 2 |
| 38 | 2MCQ | B | 2 |
| 39 | 2NAT | A | 2 |
| 40 | 2MCQ | A | 2 |
| 41 | 2MCQ | B | 2 |
| 42 | 2MCQ | A | 2 |
| 43 | 2NAT | B | 2 |

|    |      |   |   |
|----|------|---|---|
| 44 | 2MCQ | C | 2 |
| 45 | 2MCQ | A | 2 |
| 46 | 2NAT | C | 2 |
| 47 | 2NAT | B | 2 |
| 48 | 2MCQ | B | 2 |
| 49 | 2NAT | A | 2 |
| 50 | 2MSQ | B | 2 |
| 51 | 2MCQ | B | 2 |
| 52 | 2NAT | B | 2 |
| 53 | 2MSQ | B | 2 |
| 54 | 2MCQ | C | 2 |
| 55 | 2NAT | B | 2 |
| 56 | 2MCQ | C | 2 |
| 57 | 2MSQ | B | 2 |
| 58 | 2MCQ | B | 2 |

|    |      |   |   |
|----|------|---|---|
| 59 | 2MCQ | B | 2 |
| 60 | 2NAT | B | 2 |
| 61 | 2MCQ | B | 2 |
| 62 | 2MCQ | B | 2 |
| 63 | 2NAT | B | 2 |
| 64 | 2MSQ | A | 2 |
| 65 | 2NAT | A | 2 |

## DETAILED SOLUTIONS

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**Q1.**

**Answer: B**

**Solution:**

Correct Answer: B

*Subject: General Aptitude | Topic: Quantitative | Difficulty: easy*

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**Q2.**

**Answer: A**

**Solution:**

Correct Answer: A

*Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: easy*

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**Q3.**

**Answer: C**

**Solution:**

Correct Answer: C

*Subject: Physical Metallurgy | Topic: Crystal Structure | Difficulty: moderate*

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**Q4.**

**Answer: B**

**Solution:**

Correct Answer: B

*Subject: Mechanical Metallurgy | Topic: Dislocations | Difficulty: moderate*

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**Q5.**

**Answer: A**

**Solution:**

Correct Answer: A

*Subject: Extractive Metallurgy | Topic: Iron Making | Difficulty: easy*

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**Q6.**

**Answer: C**

**Solution:**

Correct Answer: C

*Subject: Phase Transformations | Topic: Phase Diagrams | Difficulty: easy*

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**Q7.**

**Answer: B**

**Solution:**

Correct Answer: B

*Subject: Heat Treatment | Topic: Steel | Difficulty: moderate*

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**Q8.**

**Answer: B**

**Solution:**

Correct Answer: B

*Subject: Corrosion | Topic: Electrochemical | Difficulty: easy*

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**Q9.**

**Answer: B**

**Solution:**

Correct Answer: B

*Subject: Welding & Joining | Topic: Arc Welding | Difficulty: difficult*

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**Q10.**

**Answer: A**

**Solution:**

Correct Answer: A

*Subject: Non-Ferrous Metals & Alloys | Topic: Aluminum | Difficulty: moderate*

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**Q11.**

**Answer: B**

**Solution:**

Lever rule: fraction of phase = opposite arm / total tie-line length.

*Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate*

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**Q12.**

**Answer: A**

**Solution:**

Eutectic: L  $\rightarrow$   $\alpha$  +  $\beta$  at eutectic temperature. One liquid to two solids.

*Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy*

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**Q13.**

**Answer: B**

**Solution:**

BCC: 8 nearest neighbors at cube corners.

*Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate*

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**Q14.**

**Answer: B**

**Solution:**

BCC: 8 nearest neighbors. Coordination number = 8.

*Subject: Material Science | Topic: Material Science | Difficulty: moderate*

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**Q15.**

**Answer: B**

**Solution:**

$f(0)=2$ ,  $f(1)=0$ ,  $f(-1)=4$ ,  $f(2)=4$ .  $f(-2)=-8+6+2=0$ . Roots at  $x=1$  (double?) and  $x=-2$ .  $f(x)=(x-1)^2(x+2)$ . Two distinct real roots.

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy*

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**Q16.**

**Answer: A**

**Solution:**

Brass: copper-zinc alloy. Bronze: copper-tin.

*Subject: Material Science | Topic: Material Science | Difficulty: moderate*

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**Q17.**

**Answer: A**

**Solution:**

Homogeneous ODE. Substituting  $y=x*v$  and  $y(1)=0$  gives  $y=0$ .

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate*

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**Q18.**

**Answer: C**

**Solution:**

FCC: 12 nearest neighbors. Coordination number = 12.

*Subject: Material Science | Topic: Material Science | Difficulty: moderate*

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**Q19.**

**Answer: A**

**Solution:**

$|\langle u, v \rangle| \leq \|u\| * \|v\|$ . Equality when  $u$  and  $v$  are linearly dependent.

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate*

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**Q20.**

**Answer: B**

**Solution:**

Hall-Heroult: electrolytic reduction of alumina ( $Al_2O_3$ ) to aluminum.

*Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate*

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**Q21.**

**Answer: B**

**Solution:**

Quenching: rapid cooling from austenitizing temp. Forms hard martensite.

*Subject: Material Science | Topic: Material Science | Difficulty: moderate*

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**Q22.**

**Answer: B**

**Solution:**

Blast furnace: iron ore + coke + limestone -> pig iron (high carbon iron).

*Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate*

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**Q23.**

**Answer: A**

**Solution:**

$L\{e^{-at}\} = 1/(s+a)$ . Here  $a=3$ :  $1/(s+3)$ .

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy*

---

**Q24.**

**Answer: C**

**Solution:**

Rolle's:  $f$  continuous on  $[a,b]$ , differentiable on  $(a,b)$ ,  $f(a)=f(b)$ . Then  $f'(c)=0$  for some  $c$  in  $(a,b)$ .

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy*

---

**Q25.**

**Answer: C**

**Solution:**

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$ . At  $\pi/4$ :  $1 + 2 = 3$ . Hmm. Let me recalc:  $d/dx \ln(\sec x) = \tan x$ .  $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$ . At  $\pi/4$ :  $1 + 2 = 3$ . Not in options. The derivative of  $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$ . At  $\pi/4$ :  $1+2=3$ . Hmm.

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate*

---

**Q26.**

**Answer: C**

**Solution:**

Kirchhoff: for  $K_n$ , spanning trees =  $n^{n-2} = 4^2 = 16$ .

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate*

---

**Q27.**

**Answer: B**

**Solution:**

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$ .

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult*

---

**Q28.**

**Answer: A**

**Solution:**

$f(x)=x$  is odd on symmetric interval. Fourier series has only sine terms.

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate*

---

**Q29.**

**Answer: A**

**Solution:**

Row 2 = 2\*Row 1, Row 3 = 3\*Row 1. All rows linearly dependent. Rank = 1.

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy*

---

**Q30.**

**Answer: A**

**Solution:**

$f(i) = i^2 + 3i + 2 = -1 + 3i$ .  $\text{Re}(f(i)) = -1$ .

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate*

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**Q31.**

**Answer: A**

**Solution:**

Standard normal:  $P(-1 < Z < 1)$  approx 0.68.

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult*

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**Q32.**

**Answer: C**

**Solution:**

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate*

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**Q33.**

**Answer: A**

**Solution:**

Poisson: mean =  $\lambda$  = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

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**Q34.**

**Answer: A**

**Solution:**

Eigenvalues of  $A^2$  are 1, 4, 9. Trace = sum =  $1+4+9 = 14$ .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

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**Q35.**

**Answer: A**

**Solution:**

Eigenvalues are 3 and 1.  $[1,0]^T$  is eigenvector for eigenvalue 1.  $A^{50}[1,0]^T = [1,0]^T$ .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

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**Q36.**

**Answer: A**

**Solution:**

PGF of Poisson( $\lambda$ ):  $G(z) = E[z^X] = \exp(\lambda(z-1))$ .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

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**Q37.**

**Answer: B**

**Solution:**

$dy/y = dx/x$ .  $\ln(y) = \ln(x) + C$ .  $y = Cx$ . Wait:  $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = Cx$ . That's option A. Hmm, let me check:  $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = Cx$ . Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

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**Q38.**

**Answer: B**

**Solution:**

$\text{proj} = ((v \cdot w)/(w \cdot w))w = (4/5)(2,1,0) = (8/5, 4/5, 0)$ .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

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**Q39.**

**Answer: A**

**Solution:**

$L\{f(t)\} = -d/ds(F(s))$ .  $F(s) = 1/(s+2)$ .  $-d/ds[1/(s+2)] = 1/(s+2)^2$ .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

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**Q40.**

**Answer: A**

**Solution:**

Area =  $\pi r^2$ . Circumference =  $2\pi r$ .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

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**Q41.**

**Answer: B**

**Solution:**

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$  as  $n \rightarrow \infty$ .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

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**Q42.**

**Answer: A**

**Solution:**

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$ .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

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**Q43.**

**Answer: B**

**Solution:**

$|x|$  is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

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**Q44.**

**Answer: C**

**Solution:**

$y = A \cos(2x) + B \sin(2x)$ .  $A=1$ ,  $B=0$ .  $y(\pi/2) = \cos(\pi) = -1$ .

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate*

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**Q45.**

**Answer: A**

**Solution:**

Eutectic: L  $\rightarrow$   $\alpha$  +  $\beta$  at eutectic temperature. One liquid to two solids.

*Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: difficult*

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**Q46.**

**Answer: C**

**Solution:**

FCC: 12 nearest neighbors. Coordination number = 12.

*Subject: Material Science | Topic: Material Science | Difficulty: easy*

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**Q47.**

**Answer: B**

**Solution:**

Hall-Heroult: electrolytic reduction of alumina ( $\text{Al}_2\text{O}_3$ ) to aluminum.

*Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate*

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**Q48.**

**Answer: B**

**Solution:**

BCC: 8 nearest neighbors. Coordination number = 8.

*Subject: Material Science | Topic: Material Science | Difficulty: moderate*

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**Q49.**

**Answer: A**

**Solution:**

Row 2 = 2\*Row 1, Row 3 = 3\*Row 1. All rows linearly dependent. Rank = 1.

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate*

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**Q50.**

**Answer: B**

**Solution:**

BCC: 8 nearest neighbors at cube corners.

*Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: difficult*

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**Q51.**

**Answer: B**

**Solution:**

Lever rule: fraction of phase = opposite arm / total tie-line length.

*Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy*

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**Q52.**

**Answer: B**

**Solution:**

Quenching: rapid cooling from austenitizing temp. Forms hard martensite.

*Subject: Material Science | Topic: Material Science | Difficulty: difficult*

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**Q53.**

**Answer: B**

**Solution:**

Hall-Heroult: electrolytic reduction of alumina ( $\text{Al}_2\text{O}_3$ ) to aluminum.

*Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: difficult*

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**Q54.**

**Answer: C**

**Solution:**

FCC: 12 nearest neighbors. Coordination number = 12.

*Subject: Material Science | Topic: Material Science | Difficulty: easy*

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**Q55.**

**Answer: B**

**Solution:**

$\text{proj} = ((v \cdot w) / (w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$ .

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult*

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**Q56.**

**Answer: C**

**Solution:**

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$ . At  $\pi/4$ :  $1 + 2 = 3$ . Hmm. Let me recalc:  $d/dx \ln(\sec x) = \tan x$ .  $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2 \csc(2x)$ . At  $\pi/4$ :  $1 + 2 = 3$ . Not in options. The derivative of  $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2 \tan x$ . At  $\pi/4$ :  $1 + 2 = 3$ . Hmm.

*Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate*

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**Q57.**

**Answer: B**

**Solution:**

BCC: 8 nearest neighbors at cube corners.

*Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate*

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**Q58.**

**Answer: B**

**Solution:**

Hall-Heroult: electrolytic reduction of alumina ( $\text{Al}_2\text{O}_3$ ) to aluminum.

*Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate*

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**Q59.**

**Answer: B**

**Solution:**

Blast furnace: iron ore + coke + limestone -> pig iron (high carbon iron).

*Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: easy*

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**Q60.**

**Answer: B**

**Solution:**

BCC: 8 nearest neighbors. Coordination number = 8.

*Subject: Material Science | Topic: Material Science | Difficulty: easy*

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**Q61.**

**Answer: B**

**Solution:**

Quenching: rapid cooling from austenitizing temp. Forms hard martensite.

*Subject: Material Science | Topic: Material Science | Difficulty: moderate*

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**Q62.**

**Answer: B**

**Solution:**

BCC: 8 nearest neighbors at cube corners.

*Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate*

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**Q63.**

**Answer: B**

**Solution:**

Blast furnace: iron ore + coke + limestone -> pig iron (high carbon iron).

*Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate*

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**Q64.**

**Answer: A**

**Solution:**

Eutectic: L -> alpha + beta at eutectic temperature. One liquid to two solids.

*Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy*

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**Q65.**

**Answer: A**

**Solution:**

Eutectic: L -> alpha + beta at eutectic temperature. One liquid to two solids.

*Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: difficult*

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## PAPER BLUEPRINT

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### Subject Distribution:

Engineering Mathematics: 29 questions, 46 marks  
Physical Metallurgy: 11 questions, 18 marks  
Material Science: 10 questions, 16 marks  
Extractive Metallurgy: 8 questions, 13 marks  
General Aptitude: 1 questions, 1 marks  
Mechanical Metallurgy: 1 questions, 1 marks  
Phase Transformations: 1 questions, 1 marks  
Heat Treatment: 1 questions, 1 marks  
Corrosion: 1 questions, 1 marks  
Welding & Joining: 1 questions, 1 marks  
Non-Ferrous Metals & Alloys: 1 questions, 1 marks

### Difficulty Distribution:

Easy: 21 | Moderate: 34 | Difficult: 10

### Statistics:

Total Questions: 65  
Total Marks: 100

*This blueprint is generated from the actual question distribution in this paper.*